



S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

PRACTICAL NO. 02

Aim: Construct a program to read the temperatures of water in degree Celsius from the user and print the appropriate output depending upon the following conditions:

- a) If the temperature is less than -273.15°C , print that the temperature is invalid because it is below absolute zero.
- b) If it is exactly -273.15°C , print that the temperature is “absolute 0”.
- c) If the temperature is between -273.15°C and 0°C , print that the temperature is below “freezing”.
- d) If it is 0°C , print that the temperature is at the “freezing point”.
- e) If it is between 0°C and 99°C , print that the temperature is in the “normal range”.
- f) If it is equals to 100°C , print that the temperature is at the “boiling point”.
- g) If it is above 100°C , print that the temperature is above the “boiling point”.

Name of Student:

Roll No.:

Semester/Year:

Academic Session:

Date of Performance:

Date of Submission:

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OBJECTIVES:

- To develop a Python program that accepts the temperature of water in degrees Celsius from the user.
- To understand and implement **conditional statements (if, elif, else)** for decision-making.
- To classify the given temperature into different states such as **absolute zero, freezing point, normal range, and boiling point** based on predefined conditions.

Theory:

Conditional statements are one of the fundamental control structures in Python. They allow a program to execute different blocks of code depending on whether a specified condition is **True** or **False**.

Python provides the following decision-making statements:

- **if** – Executes a block of code if the condition is true.
- **elif** – Checks another condition if the previous condition is false.
- **else** – Executes when none of the above conditions are true.

In this experiment, the program reads the water temperature in degrees Celsius and compares it against scientifically defined temperature limits:

- **Below -273.15°C :** Invalid temperature (below absolute zero).
- **-273.15°C :** Absolute zero.
- **Between -273.15°C and 0°C :** Below freezing.
- **0°C :** Freezing point.
- **Between 0°C and 100°C :** Normal liquid range.
- **100°C :** Boiling point.
- **Above 100°C :** Above boiling point.

A Student

(Code and output)

DISCUSSION AND VIVA VOCE:

1. What is the purpose of using **if-elif-else** statements in Python?
2. Why is **-273.15°C** considered the absolute zero temperature?
3. What data type is used to store the temperature in this program, and why?
4. What will happen if the user enters **100.5°C** as input?
5. How would you modify the program to classify temperatures in **Fahrenheit** instead of Celsius?

CONCLUSION:

This experiment demonstrates the use of **conditional statements** in Python to classify water temperatures based on scientific temperature ranges. It helps in understanding **decision-making, comparison operators, and user input handling**. The program accurately identifies whether the entered temperature is below absolute zero, at the freezing point, within the normal range, at the boiling point, or above the boiling point, thereby strengthening the understanding of control flow in Python.